

Process Control



Group 1

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Introduction

Purpose of the lab:

Use of Armfield PCT50 Level Control to maintain the setpoint after applied disturbances. Manual and software applied disturbances 

Technical nature

First part of the experiment you control the pump manually to fill and draining the tank. The second part of the experiment you use the PCT50 software to automatically fill and drain the tank.

Theory

- Adjusting valves gives a constant flow rate. Increase pump speed, increases fill-up rate in process vessel.
- A steady water level in the tank is produced when the flow into the tank equals the flow out.
- Proportional band setting always produces an offset and never reaches the set point. 
- $K_p = 100\% / \text{Proportional Band } \%$ (high P, small output changes)
- Integral term corrects the offset that remains with the P-only controller. It takes into account time it takes to reach the set point.
- $I_{out} = K_i \int_0^t e(\tau) d\tau$ (aims to reduce offset due to P-band)
- A carefully chosen P value and I value will produce a P + I controller that will reach the set point quickly without making the system unstable.

Apparatus

- Main components of the lab station:

PCT50 Level Process Software

Level indicator

Electronic pressure sensor

Outlet

Electric interface

Overflow

Variable speed centrifugal pump

Alternative size of orifice

Ball valve

Solenoid valve



Material & Supplies

- Materials and supplies used:
 - A.** Water, medium used in level control inside the vessel.
 - B.** Electricity, software PCT50 used to control the pump speed and solenoid valve.

Procedure



- Initially fill the tank
- Exercise A (manual control):
 - Adjust CV1 to create steady flow
 - Manipulate solenoid valve and pump speed manually to disturb the system and observe the responses
 - Drain the tank
- Exercise C (automatic control):
 - P only:
 - Manipulate solenoid valve and set point at different P settings
 - Observe the changes and any changes in offset
 - Choose P setting that gives stable control and small offset
 - P + I:
 - At chosen P setting manipulate solenoid valve and set point at different I settings and observe the responses. Choose an I that gets to the set point quickly without becoming unstable.

Experimental Challenges

- Difficulties encountered:
 - A. CV1 valve not sensitive enough
 - B. Human error (manual control)
 - C. Important not to miss any steps (measurement error)
- Watch out for:
 - A. Manual valve control
 - B. Steady manual pump control
 - C. Coordination between person reading the level and controlling the software.
 - D. Unstable conditions with water level oscillating above and below set point.
- **“Lessons Learned”**

Pay attention to detail and follow the procedure as directed.

Safety



- Safety Considerations:

- Electrical hazard



- Safety controls:

1. Drain the tank
2. Turn off controller
3. Shut down computer

Comment Summary

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1. 5/5
2. 95/100

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3. That's not the main objective
4. 12/15

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5. Don't paste images.
Too much text
6. 13/15

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